Sai Santhosh Gaddam

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Education

Vellore Institute of Technology (VIT) — Vellore, Tamil Nadu
*Integrated M.Tech in Software Engineering — **CGPA: 8.91/10*** — *2023 – 2028 (Expected)*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Narayana Junior College — Anantapur, Andhra Pradesh
*Intermediate (Class XII), MPC — **97.30%*** — *2021 – 2023*

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, Bash
Frontend: Next.js, React, Tailwind CSS, Vite, HTML/CSS, WebSockets, Server-Sent Events
Backend & Cloud: FastAPI, Node.js, Express, PostgreSQL, Supabase, Prisma, Docker, Vercel, REST APIs
AI / ML: PyTorch, Hugging Face Transformers, Gemini API, Whisper, Scikit-learn, Pandas, NumPy
Tools & Testing: Git, GitHub Actions, Pytest, Vitest, Playwright, Figma, Linux

Projects

RoundsAI – Inpatient Oncology Ward Copilot | GitHub | Live Demo — *Health-a-thon 2026 · Top 150*
Next.js 15, FastAPI, Python, Whisper, Gemini 2.5 Flash, Supabase, TypeScript

- Built on-device speech transcription (faster-whisper, INT8) with a 120+ term oncology phonetic normalizer, keeping all audio local at $0 per-minute cost versus cloud ASR APIs.
- Automated CTCAE v5.0 myelosuppression grading and neutropenia alerts from paper lab reports via Gemini vision OCR.
- Architected real-time bedside-to-desktop sync over Server-Sent Events; validated with 134 automated tests.

SAME-H – Proof-Bound Document Watermarking | Patent Published — *IN202641127062 A1*
Circom, SnarkJS (Groth16), Ethereum Smart Contracts, Node.js, Express, PostgreSQL, React, TypeScript

- Co-invented a document-security system that binds one Poseidon context hash across a Groth16/Circom zero-knowledge access proof, an HKDF-SHA256 watermark key and an on-chain authorization registry.
- Built leak forensics that extracts the watermark from a recovered file, matches it to on-chain commitments and returns one of five explicit verdicts, withholding attribution on ambiguous matches.
- Secured it with AES-256-GCM encryption at rest, Argon2id auth, on-chain replay/expiry checks and org-scoped RBAC.

PRPilot – AI GitHub App for Automated Code Review | GitHub — *Live GitHub App*
Next.js 16, React 19, Supabase, Gemini 3.8 Flash, Octokit, TypeScript, Vitest, Playwright

- Shipped a GitHub App that reviews PRs in 1–2 minutes, flagging security flaws with inline comments and risk scores.
- Hardened the pipeline with HMAC-SHA256 webhook verification, Supabase row-level security on every table and Zod-validated AI output to block prompt injection and XSS.
- Designed a hybrid serverless architecture (Vercel waitUntil + GitHub Actions cron fallback) with atomic Postgres rate limiting and Vitest/Playwright E2E suites.

Smart Autofill Assistant – Privacy-First Browser Extension | GitHub | Edge Add-ons — *Published · 68 Users*
TypeScript, Chromium Manifest V3, Web Crypto API, Chrome Extensions API

- Shipped a Manifest V3 extension to Microsoft Edge Add-ons, now used by 68 users, that detects and fills forms securely.
- Built a local AES-GCM vault (Web Crypto API) with zero cloud egress; blocks passwords, OTPs and payment fields.
- Added opt-in AI semantic field matching under a strict Content Security Policy that blocks remote code execution.

Experience

InAmigos Foundation (IAF) — India
Web Development Intern — *Jun 2026*

- Redesigned the foundation website's landing (hero), login and contributor pages and presented the new design to the team.
- Collected and structured data on 10 partner NGOs and built every redesign mockup in Figma.

Achievements & Certifications

- **Health-a-thon 2026:** shortlisted for Phase 2 among the Top 150 teams with RoundsAI.
- **Certifications:** AWS Cloud Practitioner Essentials · Agentic AI Certified Associate (Oracle) · Advanced Generative AI (IBM)